Changeover Tie Out Tool: Fruit Rollups Line

Business Context

Fruit Rollups system is a decoupled manufacturing system (time between processing and packaging more than 10 mins) which produces multiple flavor SKUs. It has multiple processing and packaging lines. Most of the times multiple flavors products are being produced. When producing multiple flavor products, flavours are being distributed across line (both processing and packaging). Also each line has its own production rate since many times lines are getting stopped due to failure of equipments. There is a surge belt which accumulates the processed products before packaging. When processing rates are higher than packaging rates surge belt accunulates and reversely surge accumulation decreases when packaging rates are higher than processing rates. Changing over to a new product is very challenging as it requires balancing of flavors across lines, line rates and surge levels.

Problem Statement

Fruit Rollups manufacturing system causes overuse or overpack loss during changeover to a new product due to below reasons:

- Lack of visibility of surge of any flavor
- Inefficient distributions of flavors over lines due to human operator's decisions
- Unexpected equipment failure causing lines to stop and makes difference in line rates (processing and packaging)
- Lack of knowledge on time to stop processing lines.

Waste can be minimized by optimizing the flavour distribution, line rate and time to stop processing lines.
There is need of a tool at the shop floor to help operators to control manufacturing operations better and make changeover efficient.

Document metadata

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Executive summary

A real time mathematical optimization model helps the operator to decide the optimal time to stop processing lines to reduce overpack or overuse loss.

Scope

operators are interested to know when to stop processing lines to tackle flavor imbalance across lines and surge levels. The tool will only provide them the recommendations on stoppage time with additional informations like line rates, surge levels, target production or target time to stop manufacturing.

Stakeholders & roles

Product Owner: Monojit Mandal
Business Stakeholders: Monojit Mandal
Technical Stakeholders: Monojit Mandal

Background & context

Operators in Fruit Rollups manufacturing facility reported chanllenges in operating the lines due to frequent machine failures of machines and lack of visibility in surge level data. Mainly during changeover operators faced difficulty to avoid overuse loss. If they could avoid overuse loss they could ssave 100,000 USD per year

Goals and success criteria

- Overuse Loss: 10–20% reduction
- Monetary Savings: 100,000 USD per year

System overview / Architecture

Fruit Rollups manufacturing system starts with slurry preparation which happens in drums. There are 4 drums which prepares slurry. Each drum is connected to 2 processing lines. Flavours are being added in these processing lines

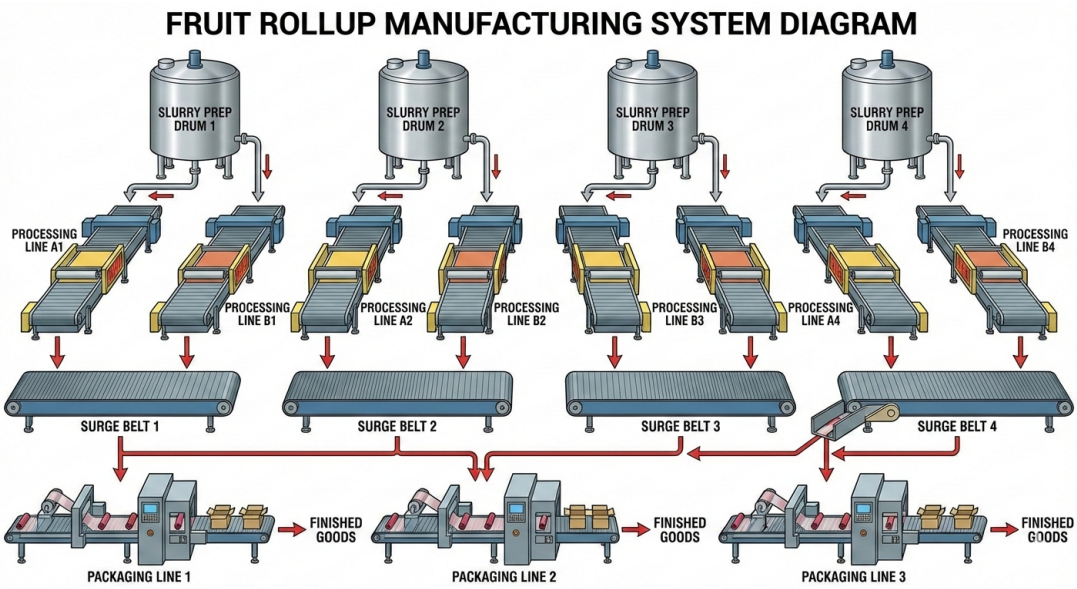


Figure 1: System topology (placeholder)

Functional components and responsibilities

For each subsystem include purpose, inputs, outputs, and acceptance criteria. Example entries:

Conveyors: Purpose – product movement and buffering. Inputs – start/stop, speed setpoint, sensors. Outputs – encoder feedback, fault signals. Acceptance – belt speed within ±5% under load.

Filling station: Purpose – dispense target volume with accuracy. Inputs – recipe, product ID, level sensors. Outputs – valve commands, dispense logs. Control – closed-loop dosing.

Capping: Purpose – apply/torque caps. Include torque checks and presence sensors.

Control strategies and sequences

Describe PLC state machines and key sequences: start/stop, emergency stop, changeover, recipe download and validation, manual/auto modes.

extbfExample (sequence):

1. Operator initiates `LineStart`; PLC verifies interlocks and safety.
2. PLC starts conveyors and brings machines to ready state.
3. PLC begins production on first product after recipe verification.
4. On fault, PLC transitions to `Fault`; operator resets and returns to `Idle` or `Running` depending on checks.

Data model and information flows

List events and payloads sent to MES/ERP/SCADA. Example fields and a simple table:

Field	Type	Descri
productionOrderId	string	MES
productCode	string	SKU
timestamp	ISO8601	event
deviceId	string	PLC
status	enum	start

Specify data retention policy (e.g., 1s raw data for 90 days, aggregated metrics for 3 years).

Interfaces and integration

Provide an interface table for each integration pair. Template:

Interface	Protocol	Payload / Format
PLC – SCADA	OPC UA / Modbus	Tag list, alarms, co
SCADA – MES	REST / MQ	Production events,
MES – ERP	REST / EDI	Material issue, conf

PLC & I/O mapping

Summarize PLC hardware and provide or link to a detailed I/O matrix (CSV or appendix). Example CSV filename: `docs/plc_io.csv`.

Safety and EHS requirements

List safety devices (light curtains, E-stops, safety PLC), how safety interacts with control logic (safe stop vs immediate stop), interlocks and LOTO procedures, and applicable standards (OSHA, ISO 13849/IEC 61508/62061 as applicable).

Alarms & HMI requirements

Define alarm severities, example alarm table, and HMI screens required: Overview, Alarms, Recipes, Maintenance, Diagnostics.

Quality and inspection

Define inspection points, vision checks, sampling plans and gating rules (e.g., reject a run after X consecutive fails). State what inspection data is logged to MES/historian.

Testing & validation plan

extbfUnit tests: function block validation and edge cases.
newline extbfFAT: functional sequences, safety tests, integration tests and performance.
newline extbfSAT: site verification, environmental checks, full integration with MES/ERP.
newline Include acceptance sign-off signature block for stakeholders.

Deployment & commissioning

Describe FAT lab requirements, rollout phases, expected downtime, rollback plans and version control practice for PLC/HMI source. Recommend storing PLC/HMI source in a VCS with tags for FAT/SAT/PROD.

Monitoring, telemetry & maintenance

KPIs to monitor: OEE, line speed, downtime, MTBF. Define dashboard needs, alerting thresholds, and routine maintenance intervals and spare parts list.

Security

State network segmentation, firewall rules, user authentication for HMI/SCADA/MES, and secure update procedures for PLC/HMI firmware and application code.

Reliability, availability & scalability

Availability targets and redundancy options (redundant PLC, dual network paths). List failure modes and mitigations.

Risk assessment & mitigations

Enumerate major risks (single vendor, supply chain, single point of failure) and proposed mitigations.

Open issues / decisions required

List decisions that need to be made (PLC vendor, MES vendor/integration pattern, mechanical layout sign-off) and owners.

Appendix & artifacts

Reference diagrams and files to attach:

- Floorplan: `diagrams/floorplan.png`
- Network topology: `diagrams/network_topology.png`
- PLC rack layout: `diagrams/plc_rack.pdf`
- I/O matrix: `docs/plc_io.csv`

Fill-in checklist

Complete before FAT:

- ☐ PLC I/O table attached

- ☐ MES interface spec attached
- ☐ Safety plan reviewed and signed
- ☐ HMI screens mock-ups provided